

Activity 09 - Conversions and Attribution Limits

C386 · v1.0 · Synthetic training data

Goal

Four synthetic conversion journeys include paid search, organic search, email and direct visits. Comparing first and last touch illustrates attribution sensitivity without pretending to reproduce GA4 data-driven attribution.

Before you start

Use a copied activity folder, Claude Cowork or Claude Code as specified, and a spreadsheet or Python3 for independent checks. No production credentials are needed. Product access varies; CSV fallback remains valid.

Mock data and assets

mock-data/touchpoints.csv

All named customers, competitors, campaigns and values are fictional. Assets include brand, source, measurement and decision contracts.

Mechanism

Define eligible channels and lookback -> order touches -> choose allocation model -> reconcile totals -> compare sensitivity -> distinguish attribution from lift.

Current Google attribution documentation supports separating allocated credit from incrementality. First/last-touch examples are authored external models, not a reproduction of GA4's data-driven model.

Worked calculation

First-touch paid search revenue = \$1200; last-touch = \$0. Email receives \$700 first-touch and \$1300 last-touch. Model choice changes assigned credit.

Step-by-step

1. Inspect touchpoints.csv and confirm four journey IDs.
2. Read assets/model-boundaries.md and define first/last touch.
3. Use Cowork Prompt 1 for channel allocations.
4. Verify each model sums to \$2400.
5. Compare paid and email allocation changes.
6. Ask Prompt 2 to challenge a recommendation to cut paid search.
7. Describe one feasible lift experiment and its outcome metric.
8. Save comparison and limitations under outputs.

Prompt1

Compare first-touch and last-touch attribution for four synthetic journeys. Reconcile each model to \$2400. Explain why this does not implement GA4 data-driven attribution or estimate incremental lift. Propose a safer budget decision.

Prompt2

Challenge the previous result. Find denominator errors, unsupported claims, bias, privacy risks and unexecuted tool boundaries. Cite exact row IDs or fields. Revise the report without inventing data.

Acceptance evidence

Both models reconcile to \$2400; paid first-touch \$1200; email last-touch \$1300; no causal ROI claim.

Failure and recovery

The agent cuts paid search because last-touch credit is zero. Assisted demand and incremental impact require separate investigation.

Use attribution for descriptive allocation; use controlled experiments or credible causal designs for incremental effects. Explicitly state identity, consent and lookback limitations.

Submit

attribution-comparison.csv and model-limitations.md

Save outputs in this working copy. Keep an execution note stating which product actions actually ran and which used fallback.